

Asset Analysis: Hydro-Pump Beta

Asset ID:	1	Assessment Date:	2026-07-13
Location:	Substation 3	Status:	APPROVED & REVIEWED

1. Executive Summary

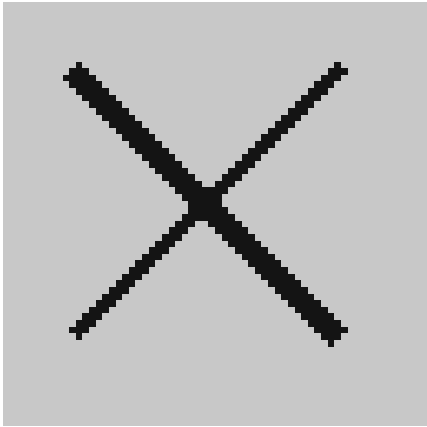
The Hydro-Pump Beta, located at Substation 3, is currently operating at a medium risk level, with diagnostic results indicating potential issues that require attention to prevent further deterioration. The asset's health status is compromised by detected cracks/scratches and notable temperature, vibration, and pressure anomalies. Overall, the pump's condition warrants proactive maintenance to mitigate potential failures and ensure continued safe operation.

2. Diagnostic Predictions

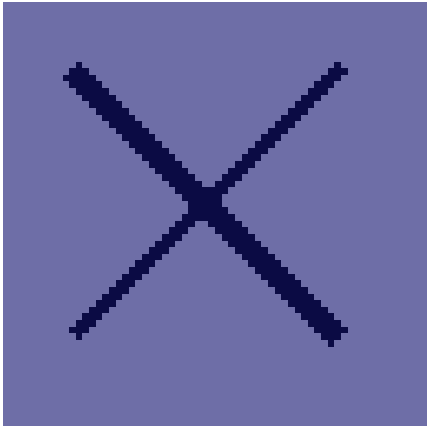
Diagnostic Module	Initial AI Prediction	Final Adjusted Status
Tabular Time-Series (LSTM)	Medium Risk	Low Risk
Visual Damage (CNN)	Damaged (Cracks/Scratches)	Healthy (No visible cracks)
Sound Analysis (ANN)	Normal Noise	Normal Noise

3. Explainable AI (XAI) Insights

Time-Series Perturbation Saliency: Temperature: 40.6% impact • Vibration: 32.3% impact • Pressure: 27.2% impact



Original Image



Visual Grad-CAM Saliency Highlight

4. Technical Risk Reasoning

The technical reasoning behind these findings is based on a combination of advanced diagnostic techniques. The LSTM time-series analysis revealed a medium risk level, with feature importance weights highlighting temperature (0.4057190716266632), vibration (0.3225538730621338), and pressure (0.271727055311203) as key factors contributing to this assessment. The CNN visual inspection detected cracks/scratches with a confidence level of 0.50, indicating a potential structural integrity issue. Although the machine sound diagnostic (ANN) reported normal sound patterns, the cumulative effect of the other diagnostic results suggests that the asset is experiencing stressors that could lead to premature failure if left unaddressed. The temperature and vibration anomalies, in particular, may be indicative of internal wear or imbalance, while the pressure anomalies could be related to the detected cracks/scratches or other issues affecting the pump's performance.

5. Recommended Mitigation & Compliance Steps

- To repair and maintain the Hydro-Pump Beta in compliance with general engineering and asset safety standards, the following action steps are proposed:
- Conduct a thorough visual inspection and perform non-destructive testing (NDT) to confirm the extent of the cracks/scratches detected by the CNN and assess the overall structural integrity of the pump.
- Implement a targeted maintenance program to address the temperature, vibration, and pressure anomalies, which may involve adjusting the pump's operating parameters, replacing worn or damaged components, or performing balancing and alignment procedures.
- Develop and implement a condition-based monitoring plan to track the pump's performance and health in real-time, using a combination of sensors and advanced diagnostic techniques to detect potential issues before they escalate into major failures, and enable proactive maintenance and repair scheduling to minimize downtime and optimize asset availability.

6. Human-in-the-Loop Analyst Verification

Analyst Remarks: Acoustic signatures and vibration perturbations verified. Overrode to low risk class based on context.

Reviewed By: Lead Inspector Sharjeel

Review Date: 2026-07-13